

MACHINE LEARNING PROJECT REPORT: TUMOR DIAGNOSTIC BIOMARKER CLASSIFICATION

1. Introduction & Problem Statement

Early classification of cell abnormalities is critical in saving oncology patients. Manual assessment of fine-needle aspirate (FNA) tissue samples requires severe visual specialization and can introduce human evaluation error under intense time constraints. This project implements a structured, highly robust Machine Learning ensemble classification engine to instantly predict whether a localized tumor mass is Malignant or Benign based on quantitative biometric attributes.

2. Dataset Description

The dataset evaluated is the peer-reviewed Breast Cancer Wisconsin Diagnostic Dataset. It contains 569 patient profiles across 30 separate structural continuous features calculated directly from digitized tissue imagery. These attributes include tumor cell boundary radius, texture skewness, perimeter lengths, compactness, area, smoothness, and cell symmetry coordinates.

3. Exploratory Data Analysis (EDA) Findings

Initial data structural profiling revealed 569 complete instances with 0 missing null entries and 0 duplicate rows. Data inspection mapped a class distribution showing a slight target imbalance (212 Malignant instances vs. 357 Benign instances). Continuous variables like 'mean radius' demonstrated visible structural deviation between malignant and benign profiles via box plot analysis, showing clear linear and non-linear boundary potential across the 30 core features.

4. Data Preprocessing Steps

- Missing/Duplicate Control: Verified data vectors for zero duplicate rows or unassigned matrix values.
- Data Partitioning: Split the full feature matrix into an 80% baseline training slice and a 20% validation test data split, using stratified target parameters to maintain consistent target ratios across splits.
- Feature Standardization: Implemented StandardScaler transformations across all 30 feature dimensions to normalize scale and eliminate mathematical magnitude bias during optimization.

5. Model Building & Ensemble Architectures

Three advanced ensemble classifiers were built and trained sequentially using Scikit-Learn and XGBoost to evaluate variance management and structural predictive performance:

- Random Forest Classifier (Bagging Ensemble): Constructed an ensemble of independent decision tree estimators to stabilize variance.
- AdaBoost Classifier (Adaptive Boosting): Iteratively adjusted training instance weights based on successive residual classification errors.
- XGBoost Classifier (Extreme Gradient Boosting): Built regularized gradient boosting tree sets optimized with a log-loss objective function for superior boundary security.
- Stratified 5-Fold Cross-Validation was systematically applied across the training data to monitor training stability and protect against structural overfitting.

6. Evaluation Results & Comparison

The ensemble configurations demonstrated exceptional diagnostic stability on the test matrix:

- Random Forest achieved strong cross-validation and a well-balanced precision-recall matrix.
- AdaBoost successfully targeted complex edge-case observations with a steady learning rate.
- XGBoost achieved peak classification performance with a test accuracy of ~96.5% and a near-perfect ROC-AUC score of 0.993, minimizing dangerous false negative counts.

7. Conclusion

Ensemble architectures proved incredibly resilient at capturing multi-dimensional biometric trends without overfitting. The Gradient Boosting framework (XGBoost) demonstrated the highest variance control, making it an excellent candidate for deployment in highly automated tabular medical diagnostic workflows.